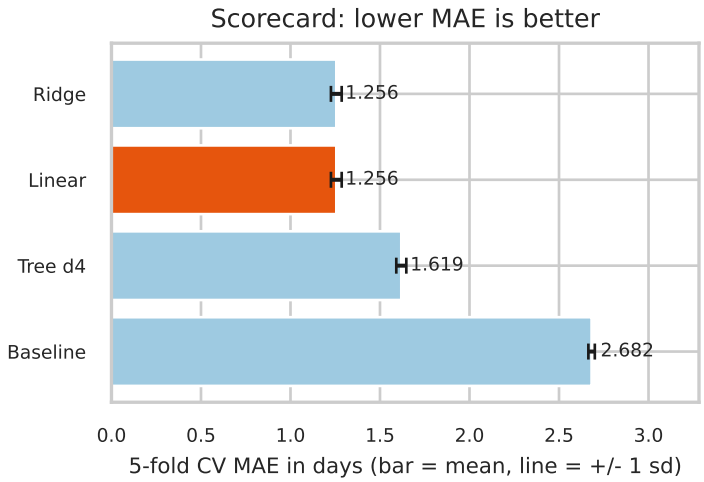
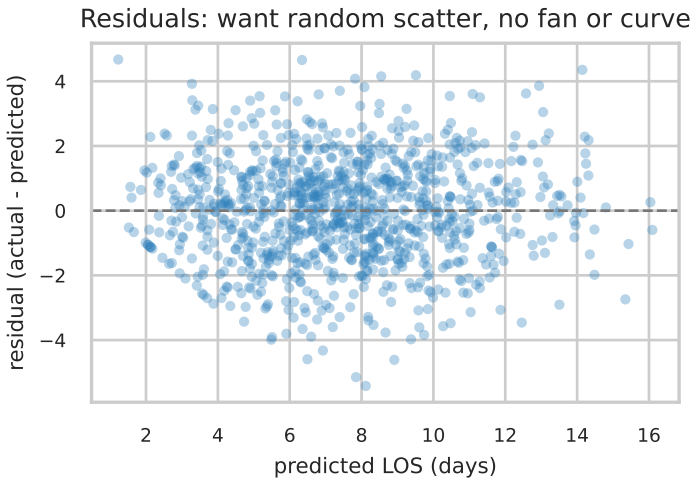
Day 22 Cheat Sheet: Predicting Length of Stay

Week 4 Day 1: regression metrics and residual checks on hospital discharges

TERM	WHAT IT MEANS	CODE
Regression	predict a continuous number (LOS days), not yes/no	DecisionTreeRegressor
DummyRegressor	mean baseline; the floor every model must beat	DummyRegressor("mean")
MAE	average |error| in days; plain-English days off	mean_absolute_error
RMSE	like MAE but squares big misses first; still in days	root_mean_squared_error
R squared	share of variance explained vs predicting the mean	r2_score / scoring="r2"
Residual	actual minus predicted; plot vs predicted	y - y_hat
Healthy residuals	random scatter around 0; no fan, no curve	ax.scatter(pred, resid)
Hold-out test	opened once, for the champion only	final.predict(X_test)
Never R2 alone	always pair with MAE and RMSE in days	report all three





THIS DATA: 4,000 simulated discharges (seed 22), 7 clinical features, continuous los_days target, 75/25 split
Mean LOS on train: 7.41 days (sd 3.31). Features never include los_days.
5-fold CV MAE: linear 1.256, Ridge 1.256, tree 1.619, baseline 2.682
CV R2: linear 0.772, Ridge 0.772, tree 0.632, baseline ~0
Champion: Linear regression. Test once: MAE 1.262 days, RMSE 1.587, R2 0.760
Within 2 days: 79% of test patients. Median |error| 1.06 days.

HABITS THAT KEEP YOU HONEST
- Include a mean baseline; if you cannot beat it, you do not have a model
- Quote MAE and RMSE in days so clinicians hear a number they can use
- R2 is useful, but never the only number on a slide or in a README
- Check residuals: fan shapes and curves mean the model is lying in a patterned way
- Write the selection rule (lowest CV MAE, prefer simpler within 1 sd) before opening test

```
cv = KFold(5, shuffle=True, random_state=22)
r = cross_validate(model, X_train, y_train, cv=cv,
        scoring={"mae":"neg_mean_absolute_error","rmse":"neg_root_mean_squared_error","r2":"r2"})
residuals = y_test - y_pred   # want mean ~0, no pattern vs predicted
```

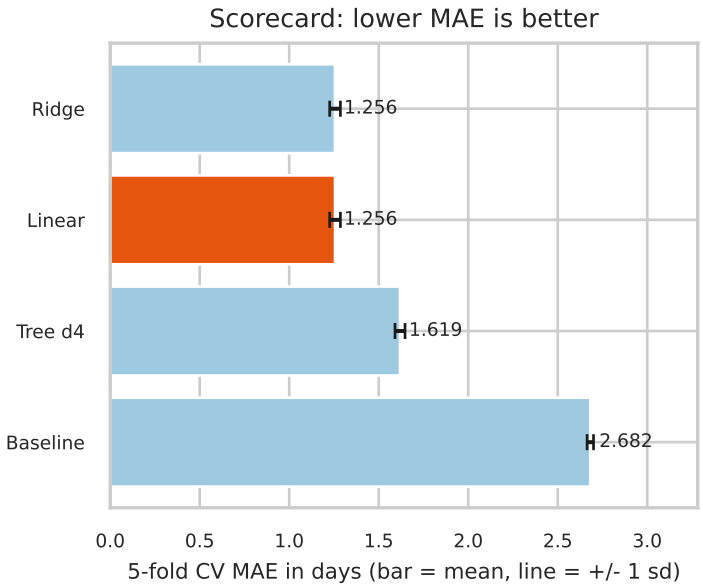
Infographic: From Classification to Regression

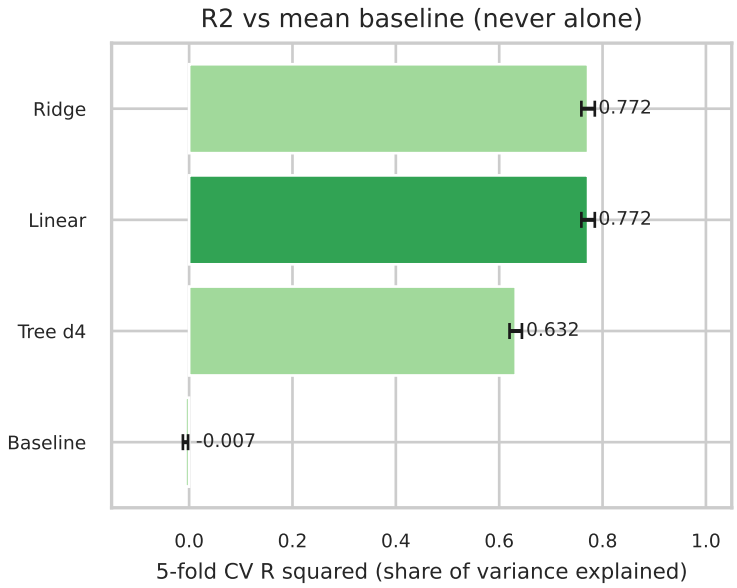
Week 4 opens: predict LOS days with MAE, RMSE, R2 and residual plots

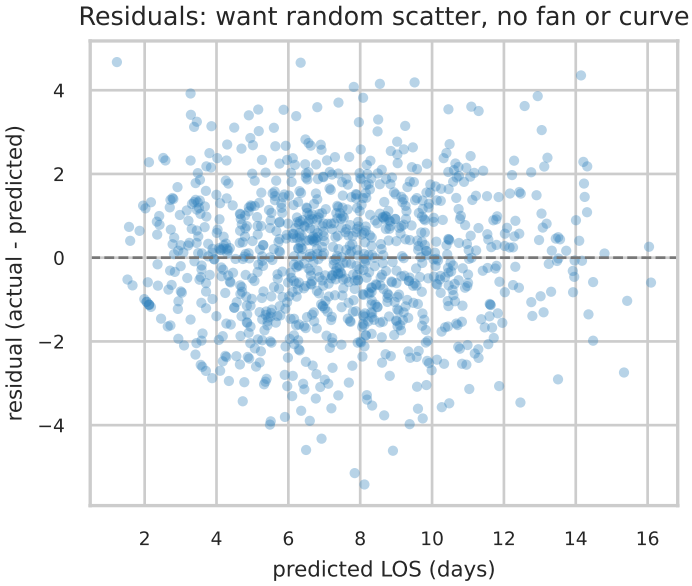
CHAMPION
Linear

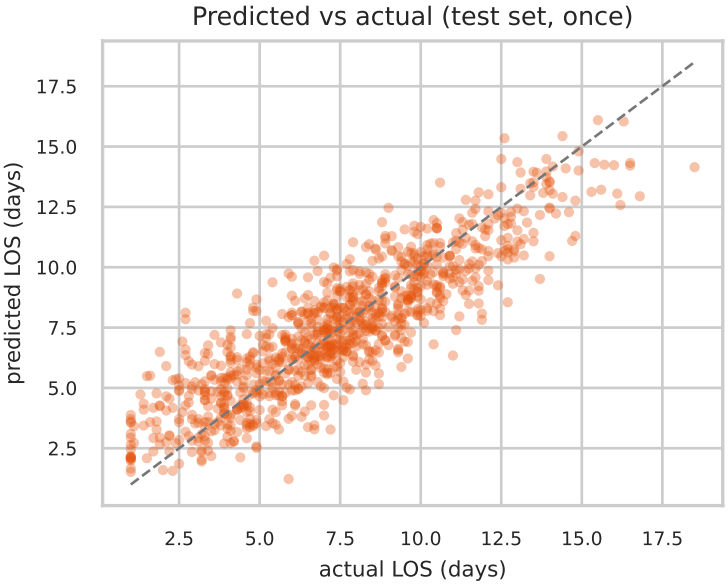
TEST MAE
1.26 d

TEST R2
0.760









Takeaways
- Week 3 was yes/no (readmission). Week 4 starts with a number (length of stay).
- MAE is average days off; RMSE punishes big misses; R2 is variance explained vs the mean.
- Linear and Ridge tie here because true LOS is a smooth add-up of clinical factors.
- Residuals should look like noise. Patterns mean you still have work to do.
- Tomorrow: regularization deeper dive (Ridge and Lasso) for correlated clinical features.

Hands-On Worksheet: ds-genai-90-days / day22

60-90 minutes | notebook: day22/los_regression.ipynb + day22/README.md

[ ] mkdir day22, conda activate ds90, open day22/los_regression.ipynb

[ ] Simulate ~4,000 discharges with continuous los_days (snippet 1, seed 22)

[ ] Confirm features do NOT include los_days; check mean/sd of target

[ ] Define models: Dummy mean, Linear+scaler, Ridge, shallow tree (snippet 2)

[ ] Run 5-fold CV with MAE, RMSE, R2 on TRAIN only (snippet 3)

[ ] Plot CV MAE bars with +/- 1 sd; note who beats the baseline

[ ] Write selection rule in markdown BEFORE touching the test set

[ ] Pick champion, score test once, print MAE/RMSE/R2 (snippet 4)

[ ] Draw residual plot (pred vs residual) and predicted-vs-actual

[ ] Do residuals look random? Any fan or curve? Note it in README

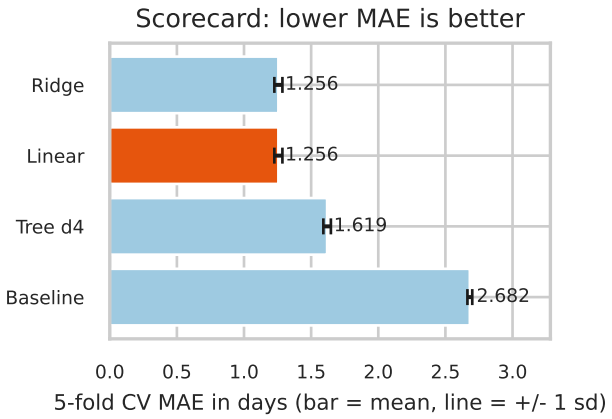
[ ] Read coefficients in days (snippet 5); which drivers matter most?

[ ] README: scorecard, two charts, champion and why, limits

[ ] git add day22 && git commit && git push (snippet 6)

[ ] Stretch: Snowflake AVG/ABS error on scored predictions (snippet 7)

```
r = cross_validate(model, X_train, y_train, cv=cv,
      scoring={"mae":"neg_mean_absolute_error","r2":"r2"})
(-r["test_mae"]).mean(), r["test_r2"].mean()
```



CV MAE linear: _____ Ridge: _____ tree: _____ baseline: _____

Champion: _____ Test MAE: _____ d RMSE: _____ R2: _____

Day 22 Self-Check Quiz

1. How is regression different from the classification work in week 3?

   _____

2. What does MAE mean in plain English for length of stay?

   _____

3. Why does RMSE punish large misses more than MAE does?

   _____

4. What does an R2 of 0.77 tell you, and what does it leave out?

   _____

5. Why must features never include los_days when predicting LOS?

   _____

6. What should a healthy residual plot look like?

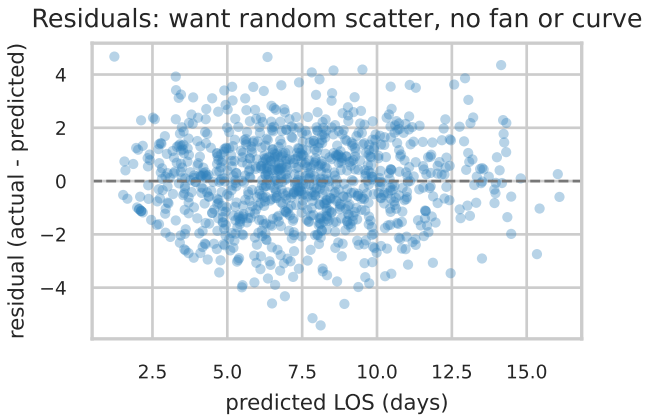
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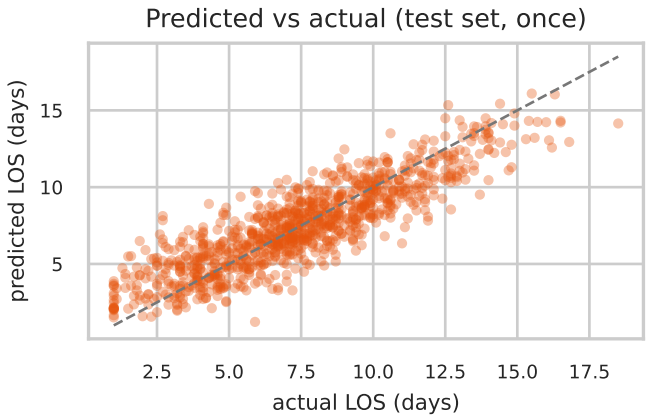
7. Why include DummyRegressor(strategy='mean') in the scorecard?

   _____

8. Why report MAE and RMSE alongside R2, never R2 alone?

   _____





Answers:
1 regression predicts a continuous number (days); classification predicts a class or probability
2 average absolute days off: here about 1.26 days on the test set for the champion
3 it squares errors before averaging, so a 6-day miss hurts more than six 1-day misses
4 the model explains 77% of LOS variance vs the mean; it does not say how many days off you are
5 that would leak the answer into the inputs; LOS is the target, not a feature
6 scatter around zero with no fan (growing spread) and no curve (missed nonlinearity)
7 it is the floor: always predicting the mean (~7.4 d); real models must beat its MAE
8 clinicians and bed planners hear days; R2 alone can hide a model that is still too wrong to use